

# Facial Expression

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## Part I — Foundations

### Preface

Subjective expectation forms the nervous system's rapid, non-deliberate judgment of impending events from all available information, typically experienced as a feeling. It follows the primitive criterion of how things ought to be and leans toward negative expectations in dark or tense settings, aiding survival while increasing the value of positive outcomes. Positive expectations require more supporting information.

Emotion ABC theory, proposed by Ellis, holds that activating event A indirectly causes emotional and behavioral consequence C; belief B, formed by cognitive evaluation of A, is the direct cause. Negative emotions arise chiefly from erroneous beliefs. Emotions precede reason in micro-reaction research. The first neural response to stimuli is emotional; rational thought follows unless the stimulus overwhelms. Emotion operates on single-layer logic: positive response to beneficial stimuli and comparative judgment leading to rejection, confrontation, avoidance, or abandonment for harmful ones. Dictionary definitions separate enduring mood states (minutes to days) from situational, transient emotions proper (rarely exceeding a few seconds except sadness), the latter marked by external manifestations and used in micro-reaction analysis.

Driving scenarios illustrate expectation mismatches. A U.S. driver in America expects rule-based alternation and registers mild contempt at violations. The same driver in China encounters non-alternating traffic, producing surprise, contempt, anger, and eventual conformity. Experienced Chinese drivers hold flexible norms and experience fewer mismatches. A Chinese driver in the United States encounters repeated mismatches at uncontrolled intersections, generating confusion, fear, and observed disgust until local rules are explained. Expectation mismatch directly generates emotion: New Year countdown silence produces strangeness; unexpected candy yields delight; network disconnection yields cursing or anger when attributed to negligence.

Emotion management requires observing feelings inwardly, expressing impact without accusation, and releasing through confiding or music while avoiding repetition of error or self-harm. Inner emotions produce

external expressions. Universal emotions are surprise, disgust, anger, fear, sadness, and happiness. Cognitive therapy, developed in the United States in the 1960s–1970s, alters maladaptive cognitions; three months produced significant improvement in 80 percent of unipolar-depression patients. Emotions follow 28-day cycles of peak, critical point, and trough. In dark environments, stimuli violating safety expectations produce intense fear; the same stimuli in bright environments yield weaker responses. Facial expressions do not necessarily indicate genuine emotion; actors can produce precise voluntary expressions. Effective stimulation tailored to the individual is required to elicit authentic emotion before micro-expressions are observed.

Terminology distinguishes an “actor” performing non-stress expressions from a “test subject” producing stress responses. The Facial Action Coding System (FACS) and its action units (AUs) provide the measurement framework. Micro-expressions last 1/25 to 1/4 second (approximately 0.2 seconds) and occur outside conscious control. Observed examples include asymmetric mouth lift during a commercialization proposal and sudden eye widening followed by pupil contraction upon presentation of a name card or photograph.

In interviews, maintain feet parallel, hands at sides or on knees, eyes level, and slight forward lean; adjust posture if the interviewer leans back. Expressions comprise static features and dynamic signals. The seven core expressions are:

- happiness (mouth corners up, crow’s feet)
- contempt (one mouth corner raised)
- disgust (nasal lines)
- anger (furrowed brows)
- sadness (inverted brows, chin wrinkles)
- fear (trembling, tense lower lids)
- surprise (wide eyes and mouth)

In daily communication, 7 percent of information travels through words, 38 percent through voice, and 55 percent through facial expressions and body posture. Charles Darwin published on universal expressions in 1872. Silvan Tomkins demonstrated the emotion-expression link in the early 1960s. Paul Ekman and Carroll E. Izard confirmed universality of six emotions across cultures. When stressed, people display comfort behaviors: men typically grab below

## Expressions as universal language

Foot movements and torso orientation reliably indicate emotional states, directing toward liked people or topics and away from disliked ones or exits. Anticipation shows in hand-rubbing or thumb-forefinger rubbing, while anti-gravity behaviors such as floating sensations mark happiness or excitement. Liars often omit personal pronouns and names.

Ekman’s team tested an isolated Papua New Guinea tribe with no Western media exposure; members identified expressions accurately, and Americans recognized the tribe’s photographed expressions. Studies of infants and blind individuals confirmed consistency. As of 2016 the seven universally recognized expressions are happiness, contempt, disgust, anger, sadness, fear, and surprise.

Bayer recorded seven deliberate expressions—anger, fear, seduction, helplessness, happiness, sadness, indifference—with viewers identifying an average of only two. Wolff found left-face changes faster and stronger, reflecting genuine emotion via the right hemisphere, while the right face conveys rational or false signals via the left hemisphere. In the Spring and Autumn period, Chunyu Kun twice read King Hui of Liang’s face and declined to speak, inferring preoccupation first with chariot driving and then with sensual pleasures.

Emotion is a short-lived, coordinated response across language, physiology, behavior, and neural systems to an appraised event, functioning as an evolved survival mechanism. Faced with negative stimuli, animals assess relative strength: weaker stimuli trigger disgust and withdrawal, equal strength triggers anger and attack, and stronger stimuli trigger fear and flight. Positive stimuli elicit joyful excitement. Human emotions arise mainly from social events processed through rules yet retain primitive logic that produces autonomic changes such as elevated heart rate, blood pressure, sweating, and muscle tension in anger.

Upper facial muscles include:

- Frontalis raises eyebrows, forming two to three wrinkles and enlarging the palpebral fissure.
- Corrugator supercilii draws eyebrows down and inward, creating vertical glabellar lines.
- Procerus pulls glabella downward, producing transverse nasal lines.
- Levator palpebrae superioris elevates the upper eyelid; its Müller's component widens the fissure under sympathetic activation.
- Orbicularis oculi closes eyelids, depresses eyebrows, and forms crow's-feet.

Physiological expressions fall into four categories: comfort (eyes close with upward roll, eyebrows lift proportionally to intensity); pain (orbicularis oculi contracts forcefully, teeth clench, upper lip rises, lower lip pulls down, mouth corners stretch horizontally via platysma); attention (furrowed brows with passive lid descent and directed gaze); and absent-mindedness (upper lids droop to pupil edge with unfocused gaze that follows head rotation).

A psychological stress micro-reaction is an inadvertent, brief, low-amplitude action after an effective stimulus that cannot be fully masked. Examples include persistent shivering and arm tightening despite verbal denial of cold, or a subordinate carrying a bedpan at arm's length while holding breath after smiling compliance. Only these unmasked traces reliably indicate true state.

In the leader–Xiao Zhang meeting, deliberate actions include the shoulder pat, chair invitation, forward lean, and verbal praise; micro-reactions appear when the leader reclines, lights a cigarette, and adopts nonchalance after hesitation, revealing an apparent promotion that is actually a demotion. At an airport, a seated businesswoman with upright posture, stiff back, motionless ankles, and clasped hands signals tension about flying or delays. Three businessmen on phones show distinct states: stiff shoulders and serious expression indicate intent to impress; legs crossed and hand behind head indicate casual speech with a familiar contact; forward lean and face covering indicate secretive action.

A 2012 Science paper by Hillel Aviezer et al. claimed body language is required to judge emotion from tennis-player photographs, but the study is flawed because athletic expressions may contain performative elements. Micro-expressions last under 1/4 second and reveal concealed emotions; only 10 percent of observers detect them. Paul Ekman's Micro-expression Training Tool (METT) trains recognition through pretest, training, practice, review, and posttest modules. Genuine expressions arise unconsciously from the seven basic emotions—happiness, anger, sadness, disgust, surprise, fear, contempt—while fake expressions follow display rules learned in childhood.

Raising the right eyebrow signals a question. Eyebrows pulled upward and tensed indicate intense fear. Eyes glancing left typically indicate recall; glancing right indicates lie construction. Sudden pupil dilation signals fear, anger, or arousal. Flared nostrils with pressed lips indicate anger. A left-corner smile is false in roughly

Microexpression foundations

Certain facial muscles resist voluntary control. During the 1982 meeting between Pope John Paul II and Jaruzelski, the Pope's maintained smile conveyed political acknowledgment. A normal person possesses 44 facial muscles capable of more than 5,000 distinct expressions.

Physiological behaviors directed solely at the self include eating, drinking, defecating, urinating, and sleeping. Only the motor system and diaphragm can be regulated by cortical will; routine diaphragm movement is managed by the autonomic nervous system via the brainstem, medulla oblongata, and pons. Polygraphs therefore measure skin conductance, heartbeat, respiration, and blood volume with high accuracy.

Micro-expressions are evolutionarily inherited, subconscious, and uncontrolled. They last as briefly as 0.04 seconds and are captured by only 10 percent of observers. They appear in national security, judicial systems, clinical medicine, and elections. Seven universal expressions occur across cultures and even among the congenitally blind: happiness, contempt, disgust, anger, sadness, fear, and surprise. Eleven further expressions—shame, humiliation, guilt, apprehension, boasting, joy, pleasure, excitement, reassurance, and satisfaction—also appear common. Micro-expressions last under 0.2 seconds when emotion is suppressed.

Six basic emotions are expressed through facial musculature. Happiness produces a smooth forehead, raised lower eyelids with crow's feet, bulging cheeks, and lifted mouth corners, with possible teeth exposure. Sadness produces a wrinkled forehead, shortened raised eyebrows forming a triangle between the brows, nasal root and eyes, slack cheeks, and a downturned mouth. Surprise, disgust, anger, and fear follow parallel morphological patterns.

Emotional behaviors follow survival logic: surprise increases sensory intake; disgust prompts withdrawal; anger mobilizes resources; fear triggers hiding or flight; sadness conserves energy; happiness discharges excitement. Variants include revulsion and contempt (disgust); irritation and sternness (anger); apprehension and anxiety (fear); sorrow and guilt (sadness); satisfaction and excitement (happiness).

Fifteen muscle groups govern expressions. Muscles 1–5 center on the orbicularis oculi (upper face); muscles 6–15 center on the orbicularis oris (lower face). The orbicularis oris is the sole unpaired facial muscle, with marginal and peripheral portions meeting at the vermilion border. Its contraction purses or protrudes the lips; opposing forces from levator labii superioris alaeque nasi, zygomaticus major, platysma, depressor anguli oris, depressor labii inferioris, and mentalis produce lip raising, corner elevation or depression, stretching, or parting. Platysma fibers pull mouth corners laterally and downward in crying or pouting. Mentalis originates below the lower lip and inserts into chin skin, bulging the chin and elevating the central lower lip chiefly in sadness. Buccinator expels air between the lips after cheek puffing. Levator labii superioris alaeque nasi dilates the nostril and elevates the upper lip. Levator labii superioris raises and everts the upper lip. Zygomaticus minor raises the upper lip in disgust and sadness. Zygomaticus major pulls the mouth corner upward and laterally. Risorius assists smiling. Depressor anguli oris and depressor labii inferioris pull the mouth corner or lower lip downward and laterally.

Head position signals include: tilted backward for pride; lowered for negation or remorse; suddenly dropped for humility; held straight for neutrality; tilted to one side with nodding for interest; repeated nodding for agreement; suddenly raised for surprise.

False expressions are revealed by facial asymmetry, timing anomalies, and sequencing mismatches. Prolonged expressions are typically false; authentic ones are brief except in extremes

## Part II — The face by region

## Brows

Facial expressions convey thoughts and emotions, intentionally or not, but must be read in clusters with gestures, posture, and speech rather than in isolation. Eyes receive the most attention in interaction, yet every feature contributes. In sign language the same hand signs for “Grandfather is ill” become a query when paired with an urgent expression.

Eyebrows act as a barometer of mood. Raising one signals doubt or incomprehension; raising both indicates strong appreciation, liking, surprise, or an impending question. Gerard Nierenberg’s experiments showed these movements reliably expose psychological states. A single brow raised while the other stays level signals surprise mixed with doubt; resolving the doubt often produces agreement. Lynn Clasen identified more than twenty distinct eyebrow dynamics, each tied to specific emotions. A slight, momentary brow raise during conversation indicates confidence. Professor Mark classified 3,000 adult eyebrow photographs and linked narrow glabellar regions to calculation and wide regions to generosity.

Angry eyebrows form a “10:10 shape”: inner ends drawn together and pressed downward while outer ends rise diagonally. This combines corrugator supercilii activity producing vertical frown lines with frontalis activity lifting the outer brows. Leveling of the outer brows signals subsiding anger. Eyebrow movements arise from frontalis, corrugator supercilii, orbicularis oculi, and depressor supercilii action. They protect the eyes and signal inner states. Anger produces simultaneous frontalis and corrugator contraction, widening eyes, lifting outer brows, flaring nostrils, and quickening breath; the head lowers and chin tucks before attack. Surprise raises brows with matching upper-eyelid elevation. Sadness lowers outer brow ends while lifting inner ends, often with orbicularis contraction that readies tears. An eyebrow flash—rapid bilateral raise and drop—signals welcome or emphasis. One brow raised indicates deliberate doubt. Knitted brows can mark worry, confusion, disgust, or thought, requiring context.

The corrugator supercilii lies at the inner eyebrows and, with the orbicularis oculi, draws the brows together to form vertical creases under strong light, worry, or anger. The depressor supercilii lowers the inner brows and creates horizontal wrinkles at the nasal root. The frontalis raises the brows and produces 2–3 forehead wrinkles in adults, widening the palpebral fissure. The procerus lowers the brows and adds horizontal lines at the nasal root. These muscles generate five primary eyebrow movements: raising, knitting, shrugging, one brow raised with the other lowered, and drawing inward while lifting outward.

- Sharply backward-sloping forehead correlates with rapid insight, inquisitiveness, and impulsiveness.
- Prominent upper forehead with flatter brow region correlates with careful reasoning, analysis, imagination, and absent-mindedness; historical examples include Franklin, Copernicus, Shakespeare, and Einstein.
- Heart-shaped forehead indicates gentleness but weak judgment; broad signals high intelligence yet frivolity; angular denotes strong character and decisiveness; narrow reflects diligence but low perseverance; rounded suggests childlike courtesy yet indecisiveness; protruding shows stubbornness; M-shaped indicates rationality and orderliness; messy hairline implies unreliable eloquence.

Straight brows are bold or stubborn; triangular, resolute yet cold; broom-shaped, late engagement or weak finish; knife-shaped, decisive but tactless; crescent, generous yet trusting; willow-leaf, honest but not family-oriented; upward-slanting, domineering; messy, physically strong but mentally simple; soft and smooth, refined, with drooping tails indicating comprehension and upward tails indicating boldness. Long brows link to mild temperament; short to irritability; thick to arrogance or strong liver qi; thin or light to delicacy or

simplicity; sparse to introversion and changeable emotion. Scattered tails signify impulsivity and financial loss. Narrow inter-brow space indicates pettiness; wide space, broad-mindedness.

Knitting the brows signals hope, astonishment, arrogance, ignorance, thoughtfulness, worry, displeasure, or difficulty. Protective knitting shields the eyes during attack, strong light, or intense emotion, lifting the cheeks while keeping the eyes open. Aggressive knitting reflects self-protection against one's own anger. Deep knitting commonly occurs with annoyance, anxiety, dejection, or indecision. Relaxed brows indicate cheerfulness after pleasant events or solved problems and increase willingness to agree to requests. Raising the brows denotes surprise, satisfaction, or happiness; a single brow raised signals doubt. Lowering the brows shows disapproval or anger. The eyebrow shrug conveys

## Eyes

Paul Broca identified Broca's area in 1861, located just behind and above the eyeball; damage produces motor aphasia. Blood flow increases here during speech, and frequent talkers develop slight downward-outward protrusion of the eyeballs. Pupils dilate four- to fivefold during excitement and constrict when calm. Blink rate more than doubles under sudden psychological pressure, as occurs with non-professional liars. Appropriate gaze occupies 30–60 percent of conversation time; exceeding 60 percent signals either attraction or impending conflict.

Three primary sensory systems appear in language and eye behavior:

- Visual: rapid speech, nasal tone, wide pitch variation, preference for visual descriptors; eyes typically move upward and left (right-handers recalling images) or right (constructing images).
- Auditory: slower, measured rhythm, careful word choice, auditory phrases such as "That sounds right."
- Kinesthetic: slowest delivery, deep voice, kinesthetic phrases such as "I haven't quite gotten hold of it"; eyes often lower or drift.

Twelve observable eye states supply diagnostic cues: calm and steady indicates a formed plan; scattered and unfocused indicates absence of a solution; sharp sideways glance signals coldness; dark and gloomy signals ruthlessness; atypical movement signals a hidden scheme; dull eyes with pale lips signal acute anxiety; blazing eyes signal anger; tranquil eyes with a smile signal pleasure; darting eyes signal boredom; steady fixation signals receptive attention; lowered eyes with tilted head signal heavy worry; upward gaze signals unwillingness to listen.

Eyelid closure is governed by the orbicularis oculi muscle; elevation of the upper lid by the levator palpebrae superioris (both under voluntary somatic control) and Müller's muscle (smooth muscle under autonomic control only). Involuntary closure occurs with negative stimuli to reduce impact. Positive stimuli also prompt closure to heighten other senses, producing relaxed expression with minimal brow or mouth change. Negative closure produces discomfort, with furrowed brows, downturned mouth corners, chin folds, and nasal wrinkles involving frontalis, corrugator supercilii, orbicularis oculi, depressor labii inferioris, and orbicularis oris. Rapid blinking arises when facing decisions.

Stanford research divided subjects by eye-turn direction during thirty-second questioning: rightward turns correlated with impatience and aggression; leftward with internalization. In the Bert prison-abuse interview, repeated rightward glances despite courteous answers led agents to provoke him, revealing his violent nature. A test subject's baseline rightward gaze shifted left only when denying dislike of a friend; follow-up image and micro-expression checks confirmed concealed disgust.

An experiment paired fifty students into twenty-five dyads for twenty seconds of silent pupil-to-pupil contact. At contact, both expected speech; after two seconds most felt awkward; after five seconds many wished to look away; at twenty seconds all reported relief and an urge to speak. People spend 30–60 percent of interaction time in mutual gaze; above 60 percent signals attraction or impending conflict. Extroverts maintain steady contact; sudden prolonged staring by normally shy individuals often indicates lying.

Incomplete closure of the orbicularis oculi occurs during genuine laughter or crying. In formal settings, superiors direct gaze at the triangular forehead zone to exert psychological dominance. Prolonged staring beyond ten seconds produces discomfort; abrupt aversion suggests evasion. Rolling the eyes from direct contact follows a sequence of upward gaze, lateral shift, and final closure, classically

## Nose

Pupils dilate involuntarily to liked or sexually interesting stimuli and constrict to disliked ones. The sphincter pupillae, under parasympathetic control, constricts the pupil; the dilator pupillae, under sympathetic control, dilates it. Both operate outside voluntary control. In one FBI sexual-harassment murder case, the suspect's pupils dilated when he mentioned the victim, contradicting his denials. Panic or excitement can expand pupils to four times normal size. Ancient Persian jewelers raised prices when buyers' pupils enlarged. Light-colored irises make dilation more visible.

Six skeletal extraocular muscles—the medial rectus, lateral rectus, superior rectus, inferior rectus, superior oblique, and inferior oblique—move each eye under voluntary control. Abnormal insertion or paralysis produces strabismus. Upward gaze shifts often indicate escape or search; downward shifts indicate endurance or guilt. Lateral shifts interrupt interaction but show no fixed link to lying.

Normal blink rate is five to eight times per minute. Several blinks within one second signal emotional arousal or shyness; a blink exceeding one second indicates boredom or contempt. Continuous blinking during speech shows divided attention; during listening it indicates feigned interest. Prolonged closure with exhalation denotes anxiety. Effortful widening signals interest, surprise, or fear. Squinting without bright light suggests uncertainty; paired with a broad smile it indicates happiness. Pleasant surprise widens the eyes, dilates the pupil, and increases scleral exposure.

The orbicularis oculi surrounds the eye and interacts with the levator palpebrae superioris, frontalis, corrugator supercilii, and zygomaticus minor. Its contraction forms crow's feet and indirectly elevates the upper lip. The levator palpebrae superioris elevates the upper lid; its relaxation with orbicularis oculi contraction produces closure. Eye baseline in neutral state exposes 71.5 % of iris vertical diameter (upper lid covers 19 %, lower lid 9.5 %). Upper scleral show correlates with self-centeredness, irritability, neuroticism, and criminal tendency; lower scleral show correlates with confidence, willfulness, materialism, and coldness.

Single-fold eyelids on both eyes correlate with introverted, sensitive, strong-willed, diligent personalities. Double-fold eyelids correlate with outgoing, emotionally rich, popular individuals. One single-fold and one double-fold correlate with low loyalty and sexual openness. Deep-set eyes correlate with rational, thorough, calm, honest, patient, responsible, persistent yet reserved thinkers. Narrow eyes correlate with meticulous, observant, orderly, kind, dedicated yet introverted, indecisive, suspicious, neurotic personalities. Round eyes correlate with easygoing, cheerful, frank, consistent, popular, quick-thinking individuals. Protruding eyes link to optimism, kindness, talkativeness, and impatience. Small eyes associate with honesty, introversion, and perseverance; large eyes with expressiveness and sensitivity.

Social gaze directed between the other person's eyes and mouth creates a relaxed atmosphere. Intimate gaze moves from eyes to chin to torso. A sideways glance with a smile conveys interest; one with a pout or lowered brows signals doubt or hostility. A woman's sidelong glance on first meeting signals interest; a man's conveys contempt. Upward gaze with lowered head projects submission and charm. Winking indicates shared

## Mouth

The NLP model claims upper-left eye movement signals visual recall and upper-right construction, with horizontal left for auditory recall and right for construction; lower left indicates internal dialogue and lower right kinesthetic sensation. A left-handed supplement reverses all directions. These assertions rely on crossed hemispheric control, yet extraocular muscles receive bilateral corticobulbar innervation, allowing only conjugate movements. Independent divergence is impossible because lateral rectus control originates in the abducens nucleus and medial rectus control in the oculomotor nucleus. A 1979 study of 200 participants answering six questions found a 63 percent average falsification rate across the six rules; only 2.6% of individuals matched the full pattern.

Blinking rate rises sharply during deception. U.S. customs inspectors observe elevated blinking after reminders about undeclared items. Pupils dilate rapidly to pleasurable stimuli, a cue ancient Persian jewel merchants used when pricing gems by customer pupil size. In terror, anger, or excitement pupils dilate; in disgust or fatigue they constrict.

Eye-roll sequences from direct contact begin with lid closure, upward gaze exposing sclera, lateral shift with head turn, downward movement, and final lid closure. From a sideways position the sequence starts with a quick glance, upward lid closure, then lateral reopening. Coquettish rolls add a final return to the interlocutor's pupils. Lateral gaze shifts interrupt mutual pupil contact but carry no fixed left-right meaning; individual baselines must be established first. Upward shifts accompany contempt, feigned innocence, or unprepared thinking. Downward shifts map to endurance under guilt, suppressed anger, or shyness.

People with uneven eyes are described as cunning strategists with volatile temperaments and fluctuating careers; they remarry more often. A man's larger left eye signals discord with father and wife plus male chauvinism; larger right eye indicates no maternal affinity and affection for other women. A woman's larger left eye predicts she will fear and serve her husband. Myopic squinting engages the orbicularis oculi muscle, yet only ciliary-muscle flattening of the lens alters true image quality.

Double eyelids mark open, flexible personalities lacking perseverance; long lids indicate emotional fragility. Single eyelids denote strong-willed, introverted, suspicious individuals. Multiple eyelids signal emotional richness. Drooping corners reflect childhood influence: good education yields benefactors; poor upbringing breeds distrust. Peach-blossom eyes denote romance and easy temptation. Triangular eyes indicate cunning aggression. Ideal eye spacing equals one eye length.

Nose-touching during lying increases because catecholamine release swells nasal cells and raises temperature via insular cortex activation, producing itch. Bill Clinton touched his nose 26 times during Lewinsky testimony, once every four minutes while lying. Pelé and George W. Bush showed the same action when concealing information. When observing nose-touching without nasal discomfort, conclude the person is excited or nervous after ruling out illness.

Nose shapes correlate with temperament:

- Aquiline: quick thinking, coldness, preference for solitude (examples: Sherlock Holmes, Tony Blair).

- Greek (straight bridge): high aesthetic taste, reserve, self-confidence.
- Long: rationality, strong self-esteem, caution.
- Short: straightforwardness, impulsiveness, cheerfulness.
- Bag (Jewish): strong desire for

## Chin

Lip-biting arises from inner tension that swells or itches the lips and serves as self-suggestion to ease anxiety, as when George W. Bush bit his lip after the September 11 attacks. Habitual biting signals self-punishment in high-self-esteem individuals facing failure. When facing accusations, the gesture conceals dissatisfaction and hints at unrevealed facts.

Mouth shapes link to traits: upturned crescents mark lively, adaptable people; downturned corners indicate caution and aloofness; straight lines reflect strong willpower; rectangular shapes denote mild, justice-oriented honesty; protruding lower lips suggest debate skill but limited advancement; protruding upper lips indicate reasonableness; elongated mouths favor sociability and reliability; odd shapes signal independence yet stubbornness.

Micro-movements arise through the orbicularis oris, depressor anguli oris, and mentalis. Tightly pressed lips show rising pressure and self-suppression; pursed lips register disagreement; upturned corners signal cheer; downturned corners signal low mood or pain; pouting expresses disbelief or contempt; tongue-licking comforts nervousness. In one negotiation, FBI observer Nicholas Bartum noted an opposing party's tightly locked lips on a disputed clause, enabling targeted discussion that resolved a multimillion-dollar shipping contract.

A rectangular smile engages only the zygomaticus major, producing an asymmetrical, slow-onset expression without orbicularis oculi activity or crow's-feet. Genuine smiles are symmetrical, rapid-onset, and slow-fade, involving both muscles. Wide, thick lips correlate with excitement over sex-related topics; narrow, thin lips correlate with composure and self-control. Among relaxed-lip individuals, over 70 percent report lack of resolve as their greatest career obstacle. Mouth corners that turn downward correlate with dejection after setbacks; upward corners appear in over 90 percent of cheerful, confident individuals. Short upper lips occur in over 85 percent of people who crave praise and lose morale under criticism.

Lip-pressing signals self-suppression and dejection, with corners turning downward; intensified pressure forms a flat line and indicates resolve or concealed secrets when eye contact is avoided. Frequent lip-biting signals self-doubt and impending concession. Unconscious mouth opening indicates boredom or low confidence. Lips pursed upward signal objections. Smiling with pressed lips produces a controlled, polite expression that typically conveys refusal; women often display it during insincere compliments. Licking the lips moistens dryness under pressure, as seen in Michael Jordan while shooting, or serves as a sexual cue. Lip-biting indicates anxiety, frustration, embarrassment, or confusion. Pouting thrusts the lower lip forward to express dissatisfaction or transitional anger. Pursing signals restrained anger. Grinding the teeth, visible as masseter contraction or canine contact, reveals suppressed anger. Covering the mouth suggests shyness, concentration, or guilt. Short-burst laughter marks cold indifference. Overlapping teeth correlate with high self-confidence, capability, arrogance, and jealousy.

Spaced teeth indicate cheerfulness, honesty, and poor secret-keeping; small teeth suggest calm endurance and sensitivity; inward-tilting teeth reflect openness to innovation and vengefulness; large teeth denote stamina and boldness but carelessness; protruding teeth signal blunt speech and self-centeredness; neatly aligned teeth mark traditional, responsible steadiness.

Hearty “ha ha ha” laughter with exposed incisors and backward head tilt occurs among lonely yet decisive people; women displaying it often hold leadership roles. Nervous laughter indicates insecurity. Whole-body shaking laughter marks open-minded sincerity. Covering the mouth during laughter points to introversion. Slight smiles exposing upper incisors signal genuine greetings among respected contacts. “O”-shaped mouths with widened eyes indicate shock. Lips twisted aside signal unease.

Key muscles include:

- Orbicularis oris surrounds the mouth 2–3 cm, closes and purses lips, and enables speech and chewing; tension produces tightening or parting.
- Levator labii superioris raises and everts the upper lip while deepening the nasolabial fold.
- Zygomaticus minor elevates the upper lip and deepens the nasolabial fold in disgust or sadness.
- Depressor labii inferioris pulls the lower lip downward and laterally, exposing lower teeth in anger, fear, or sadness.
- Platysma stretches mouth corners laterally and downward in crying or pouting.
- Depressor anguli oris pulls corners downward during grief.
- Levator anguli oris elevates corners during smiling.
- Zygomaticus major pulls corners upward and laterally to form smiles.
- Bucc

## Part III — Basic emotions on the face

### The seven basic emotions

The breakup scenario triggers surprise at an unexpected statement, followed by disgust at the ex-boyfriend, anger at the threat to the relationship, fear of irreversible loss, and final sadness once the outcome is fixed. Effective stimuli first produce surprise, then a rapid benefit-harm judgment. Positive stimuli yield pleasure of varying intensity. Negative stimuli produce disgust (rejection without threat), anger (urge to attack when threatened), fear (flight or acceptance when threat exceeds capacity), or sadness (after irreversible loss). Paul Ekman confirmed these six universal emotions—surprise, disgust, anger, fear, sadness, pleasure—through fieldwork among isolated New Guinea tribes.

Color influences emotion via natural associations. Green lowers skin temperature ~2 °C, heart rate 4–8 bpm, and calms operating-room staff and patients. Blue treats insomnia and lowers blood pressure. Yellow reduces boredom in factories and promotes cheer. Red energizes but can produce irritability. Purple stabilizes mood in pregnancy. Pink quiets violent patients in institutional settings. Blue and green calm tension and fatigue after work; red, yellow, orange, and pink lift pessimism and depression; pink, light blue, and light yellow stabilize irritability. White cools agitation; black aids insomnia. Prolonged green may induce desolation, and red should be avoided when relaxation is needed.

Emotion precedes reason in response to stimuli, mobilizes bodily energy for survival and reproduction, and is difficult to fake when intense, as in anger’s deepened breathing and accelerated circulation. Subjective expectation is the judgment the nervous system forms about imminent events from all past and present information. It registers as a rapid “feeling” rather than deliberate reasoning. In an office where colleagues avert their eyes, an assistant avoids routine tasks, and the boss draws the blinds, the resulting sense of threat arises without explicit logic. Humans generate such expectations instinctively to serve survival and reproduction. In total darkness, movement slows and limbs extend cautiously because the nervous system assumes possible harmful objects; the unknown itself is not the root of fear—only the expectation of harm is.

A blind person forms safer expectations through familiarity. Psychiatric patients who cannot cognize a threat show no fear.

The half-full, half-hungry method maintains optimal pressure. A fully fed chimpanzee ignored food outside its cage and treated the available stick as a toy; a ravenously hungry chimpanzee reached futilely, grew agitated, and abandoned the tool. Only the intermediate state prompted rational use of the stick. The same principle applies to humans: insufficient pressure produces slackness, excessive pressure exceeds limits, and moderate pressure sustains peak performance. Yerkes' inverted-U links pressure to performance: too little breeds inertia, too much produces anxiety or overload, optimal tension yields motivation.

Music therapy influences emotion through rhythm, melody, tempo, and dynamics. Melodious pieces produce cheer and stability; marches inspire enthusiasm; dirges evoke sadness. It raises cortical excitability, reduces tension, anxiety, and fear, and improves stress resilience. Beethoven's music shifts sorrow to happiness and timidity to courage. Therapy requires matching current mood (melancholic pieces for low mood) and temperament (slow adagio for irritability, bold symphonies for pessimism). An experiment with 14 post-breakup participants showed upbeat music deepened distress while desolate pieces reduced negative emotion in 12.

Emotion, per the second dictionary definition, is the experience arising from a person's attitude toward things. It links to natural needs, is situational and temporary, and shows clear external signs, unlike the more stable feeling tied to social needs. Expectation mismatch amplifies emotional impact. In a layoff announcement, relief follows when one's name is absent, dejection mixed with anger when it appears despite superior competence, and happiness when the threat is later withdrawn with a raise. In darkness, an unexpected group of friends with flowers produces strong pleasure; familiar furniture yields only relaxation; a sinister laugh plus breeze intensifies fear beyond either element alone. The mismatch is the sum of the absolute values of negative expectation and stimulus.

Emotions under study last seconds (surprise under one second) except sadness, which may persist days. They arise from expectation mismatch against a vague, often unconscious "sense that something might happen." Positive mismatches produce pleasant states; negative ones yield dissatisfaction. In a dark room, safety expectations heighten surprise or fear when violated; in bright light the same events produce milder reactions.

A 20,870-student survey across ten provinces found 70 percent wanted to confide yet only 26.73

## Part IV — Smiles

### Types of smiles

People who smile quietly tend to be introverted, observant bystanders with leadership potential. Heartily laughing individuals are straightforward, tolerant, humorous, and empathetic. Loud, unstoppable laughers value loyalty and relationships. "Hee hee" chucklers are creative, imaginative, and goal-driven. Secretive or cautious smilers are conservative, shy, and quick-tempered when standards are unmet. Those who smile only when others do are optimistic yet emotionally volatile. Downward mouth-corner smilers are stubborn, melancholic, and romantic. Mouth-coverers are gentle but calculating, distracted, and prone to overthinking. Tearful laughers are compassionate, ambitious, and helpful. Soft, flat laughers remain calm and skilled at resolving conflicts. Sharp, piercing laughers are energetic, adventurous, and loyal. Nasal laughers are modest, principled, and socially adept. Cold "heh heh heh" laughers are sinister and often derail negotiations.

Genuine Duchenne smiles last 0.5–4 seconds, are symmetrical, raise the mouth corners, and contract the orbicularis oculi so the cheekbones rise and crow's-feet form. Feigned smiles last longer than 4 seconds, are asymmetrical with the left side more pronounced, and move only the mouth corners without eye-muscle involvement. Dr. Mark Frank's research confirmed cheek elevation distinguishes authenticity. Dr. Nicholas's Carleton University study showed recorded laughter triggers emotional contagion. Gerald Zaltman states only 5 % of thoughts and emotions are conscious. In Matsumoto's study of over 100 consumers, statements and expressions were inconsistent in roughly 40 % of cases; more than half displayed micro-expressions of contempt, disgust, or anger.

- Closed-lip smile with downturned corners: fleeting, indicates stubbornness and emotional reserve.
- Slight smile showing upper teeth with eye contact: greeting among friends; linked to imagination and energy.
- Open-mouthed "ha-ha" laugh showing both rows of teeth, head tilted back: hearty laughter; associated with solitary decisiveness.
- Whole-body shaking laugh: marks open-minded sincerity.
- Responsive laugh: signals optimism and compassion.
- Exaggerated laugh from a usually quiet person: indicates deep loyalty once trust forms.
- Nervous smile that stops and resumes: reveals inferiority and self-doubt.
- Mouth-covering smile: typical of introverted gentleness.
- Fitful laugh: denotes cold indifference.

Ewan Grant termed the rectangular smile, in which lips pull away from both rows of teeth, a forced polite expression signaling discomfort. Men's laughter averages 43 Hz; women's reaches 2,083 Hz. Professor Robert Provine notes that people laugh 50 percent more often while speaking than listening and thirty times more frequently in social settings than alone. Laughter functions primarily as a social bond.

A vacant smile shows fully relaxed levator palpebrae superioris, frontalis contraction lifting brows, and minimal cheek elevation. In Japanese and other Asian contexts an emotional reflex substitutes a social smile for anger, contempt, sadness, fear, or disgust. Mixed expressions are identified by unilateral mouth raise for contempt, brow tension for anger, inner brow raise for sadness, nasal wrinkles for disgust, and raised and drawn-together brows with tensed lower lids for fear.

Two muscle groups govern smiles. Zygomaticus major, under voluntary control, retracts the mouth corners, exposes teeth, and lifts cheeks. Orbicularis oculi muscles around the eyes contract involuntarily, narrowing the eyelids and producing crow's-feet. A real smile therefore begins at the eyes and lasts 2–4 seconds; a fake smile moves only the mouth, may appear asymmetrical, and can persist far longer. Genuine smiles are symmetrical, rapid-onset, slow-offset, and engage eye wrinkles; feigned smiles are slower and asymmetrical. Loud laughter mobilizes 53 facial muscles.

Pleasure arises when a stimulus positively exceeds subjective expectations. The four classic joys illustrate sudden relief from low expectation: long drought followed by rain, encountering an old friend abroad, outperforming peers on examination systems that target normal distributions, and "better than below" humor such as the xiangsheng piece *Learning to Dance*.

Paul Ekman's experiment exposed students to a horror film presented as heart-warming; those who avoided correcting him produced asymmetrical smiles lacking orbicularis oculi contraction. In micro-reaction testing, supplying praise that exceeds expectation elicits authentic smiles. The 2011 chamber-of-commerce president case showed forced amplitude mismatch plus flickering gaze indicating

## Part V — Microexpressions

### Emotion microexpressions

Sigmund Freud stated that one person's unconscious can influence another's, producing body-language judgments. Harvard researchers Robert Rosenthal and Nalini Ambady found that undergraduates' ratings of 32 silent teacher video clips matched actual end-of-semester student evaluations. Alexis Carrel noted in *Man, the Unknown* that faces reveal inner states. August B. Hollingshead's viewers correctly identified only two of six acted emotions on average.

Comfort responses unrelated to speech include nose-rubbing, scalp-scratching, hand-wringing, gum-chewing, and reaching for a drink. Freeze responses reduce movement amplitude and frequency while contracting posture, with shoulders shrugged, spine rounded, and knees and ankles pressed together. Respiratory amplitude or rate increases signal rising arousal; deliberate slow deep breathing indicates guarded opposition. Brief humming or whistling after negative stimuli marks auditory comfort. Victory reactions feature frequent eyebrow lifts, irrepressible smiles, lively shifting eyes, head swaying, rapid shoulder shaking, torso rocking, finger tapping, and swaying crossed legs. Escape reactions turn the face away with tension. Surprise produces a brief head lift, widened eyes, raised eyebrows, and slightly open mouth.

After escape, gaze stabilizes: upward signals passive avoidance; downward signals restrained avoidance with attack potential; direct contact signals confrontation. Upward gaze strengthens mindset and confrontational intent, shown by increased jawline-cervical spine angle or torso straightening. Humans inherit this from larger animals that win fights by height changes, though actual height gain remains negligible. The brain uses 20–25 % of body oxygen despite comprising 2 % of weight; insufficient supply impairs focus and raises irritability. Deep breathing counters excitement by increasing oxygen for clearer control.

Approach reactions express positive intent through tentative low-amplitude movements when outcomes are uncertain or vigorous high-amplitude actions once judged favorable. Escape and approach divide love-hate reactions by distance: liking or interest draws closer; dislike or fear pushes away. Negative upward gaze signals guilt, sadness, or powerlessness with weak mindset, shown by spine bending and head lowering. Failure reactions display mental listlessness and physical sinking, with spine unstraightened and limbs inactive. Combat reactions progress through preparation (self-directed force such as fist clenching or teeth gritting), power release (overt force on objects such as table slapping), and defensive protection (torso bending with barriers shielding chest and abdomen).

Genuine expressions on disguised faces last fractions of a second, as when Clinton's forehead furrowed briefly while mentioning Lewinsky. Victory reactions reflect excitement from perceived gain; failure shows dejection from irreversible loss. Complex smiles blend features:

- Surprise smile: raised brows and upper eyelids exposing iris, U-shaped lower lip, cheek grooves.
- Expectant smile: surprise brows and eyelids without full mouth coordination.
- Disdainful smile: one-sided upper-lip raise with nasolabial folds, closed eyelids lacking smile grooves.
- Angry smile: furrowed brows, tightened lower lids, teeth biting lip while mouth corners lift.
- Fearful smile: flattened brows, wrinkled upper lids, bulging lower lids with smile grooves.
- Sad smile: raised inner brows, tightly closed eyes, lower-lid smile shapes.
- Cunning smile: lowered furrowed brows, narrowed eyes, lips pressed with chin bulge.
- Distracted smile: relaxed drooping upper lids, faint mouth-corner lift.
- Smug smile: unraised upper lids, gaze shifted away, suppressed mouth with nasolabial folds.

Blushing arises from sympathetic activation or adrenal secretion during guilt, shame, anger, nervousness, or excitement. Women blush more readily. Extreme anger alternates red and pale via vasoconstriction. Interlaced fingers convey varied meanings by placement and tension: natural placement signals confidence; tight clasping indicates anxiety or denial; position between chest and abdomen expresses refusal; resting on thighs with moving thumbs shows dilemma; one thumb extended upward denotes certainty; fixed gaze signals suppressed aversion; propped before the face indicates hostility; rubbing produces self-soothing.

Nodding or shaking deviates from simple affirmation or negation. Slow repeated nods show interest; rapid nods signal impatience or agreement; slow shakes indicate disagreement; rapid shakes convey strong negation or, when small with lowered head, shyness. Out-of-sync nods suggest inattention. Expressionless faces may mask turmoil, producing "dead-mask syndrome" with odd complexion, blinking, nose wrinkling, or twitching. Tolstoy identified 97 smiles and 85 gazes. Micro-expressions last as briefly as 0.04 seconds and under 0.2 seconds

## Training and practice

Psychological testing reveals no fear expressions absent direct physical threat, with worry the most common micro-expression. In the Maxine Kenny case, anger converted to fear or sadness after her daughter Gail's rape and murder by David Lynn Scott III: initial anger during the witnessed violation shifted once resistance failed, with grief dominating after death confirmation. Ekman lists anger variants including restraint, intentional violation, frustration, insult, rejection, and offense to beliefs, all reducible to threats against interests. A 2011 Weibo vote of 931 participants found 72 % selected "Why?! Why did you hurt her?!" over "She's already dead!" as the muttered response.

Loss produces sadness. Ekman divides anger into themes such as infants having freedom restrained, intentional violation, frustration or powerlessness, insult, rejection by a lover, others' anger, disappointment especially in loved ones, and offense to beliefs or behaviors one opposes. He summarized frustration, others' anger, threat of harm, and rejection as variants of infringement on the subject. The root stimulus is any threat to one's interests.

David Lynn Scott III was arrested a year after the crime and convicted after four more years. On execution day Maxine addressed the offender: "You think you're a samurai? Wake up! This isn't feudal Japan... You raped and murdered because you mistakenly believed that identity gave you such a right. You raped my daughter Gail, abused her, and brutally killed her—not once, but seven times. Her hands were covered with defensive wounds." Hypothetical sequences include: witnessing the violation produces dominant anger to protect the child; violent attack on Maxine yields anger while fight outcome remains uncertain, then fear and sadness once powerlessness sets in; immediate departure after the killing leaves inconsolable grief once death is confirmed; reappearance after confirmation re-triggers intense anger.

Tightly closed lips combined with the standard anger eyebrow-eye configuration form the breath-holding expression. Anger energy engages three muscle bundles at the mouth: levator labii superioris and levator labii superioris alaeque nasi raise the upper lip and nasal wings, producing nasolabial grooves; orbicularis oris and mentalis press the lips together, thinning them into a straight line with the lower lip protruding; depressor anguli oris pulls the corners severely downward. This "breath-holding mouth" is difficult to fake because it requires coordinated action of three muscles and must accompany the standard anger eyebrow-eye morphology. Muscle features include corrugator supercilii contracted, orbicularis oculi contracted, levator palpebrae superioris contracted, orbicularis oris contracted, depressor anguli oris contracted, and mentalis contracted.

Eye-brow-eye morphology shows brows strongly pressed downward, upper and lower eyelids tensed, and gaze focused on the stimulus. Greater intensity shortens the distance between brows and upper eyelid margin. Levator palpebrae superioris contraction widens the eyes and accumulates skin folds. When gaze is unfocused, lowered brows and pressed lips indicate concentration rather than anger. Micro-expressions favor anger recognition; sustained movements favor thinking.

Driving readily elicits anger through expectation mismatches such as failure to accelerate or un-signaled lane changes. When behavior constitutes malicious provocation it registers as a dignity challenge. Urgent post-arrival goals increase anger likelihood; aimless cruising does not. Being struck triggers anger mainly through ensuing insurance, repair, and schedule disruption. During World War II, photographer Yousef Karsh removed Churchill's cigar, producing the classic "bulldog" configuration of lowered eyebrows, jutting lower lip, and tense lower eyelids.

Anger arises when an interest threat to survival or reproduction is judged eliminable by one's own action. Evenly matched adversaries or challenges to territory or competence activate the response. Overwhelming threat substitutes fear; dismissed threat substitutes disgust. In a test interview, amateur inventor Xiao Lu displayed full anger expressions only when his core self-justification—building inventions to evade ordinary responsibilities—was challenged.

Typical morphological features comprise eyebrows lowered and pressed toward the upper eyelid margins, upper eyelids raised to expose more iris, lower eyelids tensed and straightened into a penetrating glare, and mouth variants including bared teeth, pressed thin lips, or corners pulled down. Full anger engages levator labii superioris and levator labii superioris alaeque nasi to lift the alae and flare nostrils, orbicularis oris and depressor anguli oris to press lips together, and mentalis to bulge

## Part VI — Deception

### Foundations of lie detection

Keeping an appointment tests character and trustworthiness, as time is non-renewable. Vanderbilt scheduled a 10 a.m. meeting; the young man arrived at 10:20 and lost the railroad position. PepsiCo president Carl Weilerup exceeded a 15-minute appointment with merchant Jack, who placed a reminder card on stage; Weilerup then honored the meeting and extended it to 30 minutes, building lasting trust. Leaders test subordinates with open tasks or closed deadlines to reveal responsibility. Key procedures include stating exact times, confirming capability, and distinguishing objective from subjective delays.

Personal charisma draws collaborators. Jack Ma led Alibaba's "Eighteen Arhats," raised initial capital from 500,000 yuan to a 500-million-yuan stake in five years, and attracted partners through integrity. He described himself as unattractive yet effective. His wife Zhang Ying cited his founding of Hangzhou's first English corner and top-teacher ranking. Napoleon Hill observed that wealth-attracting personality moves even cold hearts.

Thorough planning converts time into money. A leader who wasted half of a three-hour report period on tea, calls, and chatter used daily time logs for two weeks to eliminate idle talk and complete work on schedule. Dr. Robert Schuller's 1968 crystal-cathedral plan listed ten donation tiers—from one 7-million-dollar gift down to 10,000 windows at 700 dollars each—plus 140 gifts of 50,000 dollars. Within 90 days he secured two 1-million-dollar donations; 10,000 windows sold in six months, and the 10,000-seat cathedral opened in 1980.

Willingness to accept losses builds long-term partnerships. Li Ka-shing routinely settled for six-tenths when eight-tenths was possible, ceding two-tenths to partners; with 100 such alliances his total exceeded what

fewer partners at eight-tenths would yield. His son Li Tzar-kuoi recalled the principle of treating others generously. Song Chengrui flew overnight from Tongchuan to Hangzhou and Guangdong to source a 300-yuan electrical component for a client whose factory lost 50,000 yuan daily without it. He spent over 4,000 yuan on travel yet delivered it the next day, earning widespread trust.

People who value knowledge feel proud of their profession. National hero Xu Zhenchao mastered "silent operation" and "one-hook accuracy," unloading 40 toxic-chemical containers onto a train in 90 minutes without sound or error. "Tsinghua's steamed-bun god" Zhang Liyong scored 650 on TOEFL through continued study.

Drunk traits include: singing (optimistic, orderly lives); arguing or fighting (emotionally unstable); immediate sleep (rational, self-disciplined); reckless promises (vain, timid); weeping or confiding (inferiority); finger-guessing games (lonely); laughing (easygoing, humorous).

Inform someone of confidential matters and test whether the secret holds. Receptionist Liu Jing mentioned clients at dinner, derailing a project. Employee Wang Qiang told Zhao Yong of an impending promotion; Zhao reported it to block him. Let a candidate handle money and observe honesty. After a prior cashier kept client gifts, foreign-trade manager Zhao Gang placed probationary cashier Liu Fang in charge of an uncounted batch of red porcelain cups. Liu Fang bought archive folders for 0.6 yuan each after Zhao had confirmed a 0.3-yuan wholesale price and later failed to account for three missing cups. People facing money fall into three categories: those who openly demand it, those who feign rejection while secretly taking it, and the rare few who remain upright.

Assigning a difficult task reveals ability. Li Yan failed to deliver grain on schedule during Zhuge Liang's sixth northern expedition and was dismissed; Xiao He kept Liu Bang's supply lines open and became chancellor. Ma Jiandong sold 476 vehicles in five months, rose to sales director of a firm with 107,000 vehicles sold and net assets over 100 million yuan, and earns 600,000 yuan annually.

Joe Navarro notes that 60–80 percent of communication is nonverbal; cuff or watch adjustment signals internal wavering. Suspects unable to recount events in reverse chronological order are lying. Men tugging collars often do so because lying produces an itchy neck sensation. Yu Pengnian donated more than 25 million yuan to Hunan projects, invested 1.8 billion yuan in the Pengnian Building

## Eyes and mouth in deception

FBI interviewers read downward gaze, turned-away face, or pressed lips as loss of interest, while raised chin, relaxed mouth, and visible teeth signal acceptance. Sustained mutual eye contact of five seconds or longer, incidental arm or shoulder touches, and head tilt indicate receptivity. An ear touch or earlobe pull shows continued interest; folding the pinna forward or inserting a finger into the ear signals desire to end the topic.

Collar tugging arises from irritation-induced heat or neck-nerve itch during deception. Women more often avert the face or change the subject, accompanying a false smile that lifts only the mouth corners without periorbital lines. Repeated cuff or watch adjustment marks internal wavering. Joe Navarro states that 60–80 percent of communication is nonverbal; agents press the topic each time the gesture appears.

Requiring suspects to recount events in reverse chronological order exposes fabrication. Lying appears in faster, smoother speech; evasive answers with moving mouth but still hands; long pauses, rigid posture, minimal movement; avoided eye contact or enlarged pupils; and markedly less fluent speech from introverts.

Neck scratching—index finger on the side of the neck or below the earlobe, roughly five times—signals uncertainty or rejection. The Brokaw Trap occurs when verbal clues are over-weighted without baseline comparison. fMRI and polygraph tests compare physiological responses, as when only the correct stolen amount (5,000 yuan) produces stable fluctuations.

Micro-reaction analysis detects contradictions, such as saying “It doesn’t hurt” after dropping a rock on the foot while clenching fists. In the York–Charlie rape case, repeated “we” and an undamaged chain indicated prior acquaintance. Occasional liars avoid names, avert eyes, raise vocal pitch, display fleeting micro-expressions, touch the nose or collar, and cover the mouth.

Anchors form by pairing neutral stimuli with rising positive emotion. Music preferences correlate with traits: symphony listeners are expressive yet trusting; rock fans seek stimulation and feel alienated; pop listeners favor simplicity; classical listeners are reflective. Ballet signals patience, tap dancing signals energy, waltz signals calm refinement.

Visually dominant people walk and speak quickly, breathe high in the chest, and look up-left when recalling. Auditory-dominant people speak slowly, tilt the head when listening, and gesture at mid-torso. Kinesthetic-dominant people favor touch, breathe diaphragmatically, and gesture near the abdomen. Anchors are set only while pleasant emotion ascends.

Psychologist Feldman reports people lie at least twenty-five times daily. Controlled experiments show lengthened response times, increased eye movements, and inappropriate laughter when lying. Normal blink rate is 30–50 per minute; rates rise under lie-induced pressure. Clinton touched his nose 26 times while testifying.

Genuine Duchenne smiles engage involuntary eye muscles producing crow’s feet; voluntary zygomatic smiles lack them. Supervisor examples showed absent crow’s feet indicated false agreement with Xiao Qing,

## Part VII — Body language

### Hands

Courtship between men and women relies on instinctive body signals. Women initiate in 90 percent of cases and follow five steps: sustained gaze until eye contact, fleeting smile, grooming, closer conversation, and incidental touch. Thirteen common female signals include head tilt with hair play, moist or parted lips, self-touching, inner-wrist display, stroking cylindrical objects, raised shoulders with sidelong glance, hip sway, pelvic twist, handbag placement near the man, one leg tucked with knee pointed toward interest, shoe dangling, and deliberate leg pressing or crossing-uncrossing.

Men emphasize power and status through thumbs hooked in belt, crotch exposure while seated, body and toe orientation toward the woman, prolonged gaze, and frequent crotch adjustment. Head lowering with upright spine signals social submission but is often false unless paired with moderate spinal curvature indicating genuine acceptance. Pronounced lumbar curvature plus limb flexion denotes extreme fear or flattery. Standard apology bowing requires observed lumbar curvature after unexpected stimulus to confirm remorse. Body drooping—vacant eyes, gravity-driven trunk and head drop, bent legs, shallow breathing—marks psychological defeat or sadness, as seen in the high-quality single woman’s shift from grief to silent collapse on a sofa. Pride appears as energetic posture, forward lean, lifted chin, and “leisurely fingers” drumming or straight-leg foot wag.

In one observed exchange, the car model S held professional posture with figure-four leg cross, upright torso, and folded hands. Light topics tilted her crossed leg slightly. Sharp personal questions triggered widened stares, nostril flaring, aborted arm-crossing, and rising lower leg to vertical. Intense aggression ended her smile; she leaned back fully, showing disgust-driven escape. Approach responses require torso-to-torso contact plus active arm pull; insincere embraces show shoulder-only touch, angled torsos, and minimal arm tension. Arrogance displays raised chin, backward-tilted head, and half-lidded downward gaze, as exhibited by Takayama Kiyoshi in his 18 November 2010 arrest photo. Surprise widens eyes and parts lips, often masked by immediate brow furrow.

Weakened breathing occurs in depression or fear via reduced amplitude; accessory muscles include sternocleidomastoid, platysma, scalene, latissimus dorsi, and pectorals. Leg or foot contact signals emerging sexual intimacy; withdrawal or freezing on touch indicates aversion. The diaphragm is a thin sheet of skeletal muscle separating the thoracic and abdominal cavities and driving respiration.

In a bribery investigation, a technical specialist with a doctorate from University X in internet security displayed repeated territory-freeze-territory-freeze posture shifts. When asked about position and history he sat erect, legs at 90 degrees, hands on knees. On technical questions about Project Z he maintained expansive posture and fluent speech. When asked about modifying specifications or contact with Mr. A he crossed legs, encircled knees, contracted torso, and bent forward. These shifts preceded confession and written admission of altering bidding documents for cash.

Anything perceived as larger than oneself elicits upward gaze. Ancient survival pressures from large carnivores, tall hunters, and natural formations imprinted this response; later cultural expressions include oversized imperial palaces, houses, cars, and religious statues such as the Leshan Giant Buddha and Bamiyan Buddhas. Freezing occurs when prey detects motion cues via air, scent, sound, or sight. A gazelle halts, raises its head, and evaluates threat type, number, speed, distance, and escape routes within roughly one second. Frogs detect only moving prey and starve among stationary flies.

Eighty percent of communication occurs through body language. Disgust-based evasion is slow and frontal, featuring torso lean-back against chair with stiff legs and face. Fear-based evasion increases head-to-stimulus distance while support points stay fixed. Dominant individuals move freely inside their territory; subordinates may not. Qing emperors granted "riding horseback inside the Forbidden City" or "sedan chair inside the Forbidden City" as highest honors. Long-term defeat produces habitual lowered chin, upturned eyes, slight forward hunch, contracted limbs, and constrained movement. A top-institution doctoral student maintained this posture despite outward success; testing revealed repeated romantic rejections driving compensatory over-discipline.

Mouth-blocking reactions attempt to conceal already-visible expressions. Covering the mouth while laughing follows three steps: genuine amusement, recognition that open laughter is inappropriate, then hand placement to signal restraint. The same blocking occurs when crying. When speaking, mouth covering before words signals unwillingness

## Arms and shoulders

Communication carries emotion through tone, volume, speed, and manner. Negative lawyer actions that alienate jurors include lack of eye contact, over-reliance on transcripts, crossed arms, hands in pockets, and stiff motionless posture. Juror doubt appears as crossed arms or legs, expressionless or unhappy faces, tightly clasped hands, ear scratching, eye rubbing, critical pondering posture, or furrowed brows. Acceptance shows in open arms or legs, Lincoln-style seating, or chin-supported or chin-stroking postures.

Universal physiological principles apply across societies, races, ages, genders, and education levels by treating the person as an animal. A Washington patrol detective stopped an armored-car robbery after noting a woman's scarf, slow approach, tightly drawn lips, furrowed brows, tense eyes, gloves, and drawn pistol; she belonged to a terrorist group. Real-estate agent Xiao Nie read a client's furrowed brows, tissue-wiped chair, sideways seating, and clutched handbag as signs of a withdrawn, suspicious, neat-freak seller who called back within two hours. Colleague Xiao Juan noted Director Wang's sweat, frown, and downturned mouth as a terrible mood and later recognized Xiao Chen's bloodshot eyes, hair-raking, yawning, and irritability from late-night work.

Former FBI agent Joe Navarro flagged French negotiators' lip tightening on key clauses, saving British shipbuilders millions. Mental models, coined by Kenneth Craik, are embedded assumptions shaping behavior. Peter Senge listed five traits: universality, concealment, duality between espoused theory and practice, and parochialism that excludes contradictory information.

Gestures of doubt, uncertainty, concealment, or suspicion vary in weight by context and are sometimes called "left-handed." A colleague's repeated left-hand face touch during sensitive topics revealed opposition once statements were rephrased as questions. Avoiding eye contact signals insincerity, though excessive contact can also indicate deception. Rejection postures include crossed arms, sideways turns with sidelong glances, or peering over glasses. Classic doubt signals are index-finger nose rubbing, ear rubbing, or eye rubbing. Professor Ray L. Birdwhistell caught a lie when a man lightly rubbed his nose while claiming to like a book. Negotiators monitor these touches; frequency drops as agreement nears, while uncrossed arms and forward lean signal narrowing distance.

Behavioral psychology, founded by John Watson, infers mental states from conditioned reflexes. An FBI agent identified a suspected spy when an index finger twitched at the name "Rooney Smith." Old Niu's teacup placement beside Old Xu's produced a frown and cup lift revealing growing distance. FBI agent Navarro located a fugitive after a mother repeatedly touched her neck while denying the son's presence. A murder suspect's 0.2–0.4-second eyelid droop at the mention of an ice pick exposed the weapon. Sigmund Freud observed that sealed lips still allow fingertips and pores to betray secrets.

Freud's iceberg model places conscious awareness at the visible tip while 95 percent of personality lies submerged in the subconscious. Dr. Konstantin estimated the brain holds roughly 150 billion neurons, at least 95 percent undeveloped; Einstein used only about 2 percent of subconscious capacity. Interviewee Meihong ended a session early after an executive repeatedly checked her watch and angled feet toward the door, then walked her to the parking lot and received a job offer the next morning.

Confident speakers form a steepled-hands gesture. FBI agents note avoidance of face touching, upright posture, and sustained eye contact. Steve McQueen learned from Henry H. Calero that his character's steeple in *The Cincinnati Kid* revealed a winning hand. The upright steeple signals confidence or superiority; the lower version, more common in women, appears while listening. An inverted steeple after positive signals indicates acceptance; after negative signals it indicates rejection. Resting the chin on one hand with a tilted head signals genuine contemplation, as in Rodin's *The Thinker*; the same posture with drooping eyelids signals boredom. Additional cues include chin or beard stroking and sliding glasses down the nose.

Emotional sweating occurs on palms, backs of hands, face, neck, forearms, and lower legs. A healthy person's palms stay dry during general perspiration; frequent palmar sweating pairs with cold, clammy hands. Intense stress can produce full-body sweating, cold sweats, pallor, visual disturbances, weakness, or fainting. Hand tremors are either physiological (small, rapid, irregular, triggered by tension,

## Legs and feet

The wood-shaped hand is flat, long, moderately thick, firm, elastic, bluish-dark, with prominent veins, muscles, and knuckles, long slender knotted fingers, and a strong upright thumb. Owners are independent, stubborn, soft-hearted, and excel in religion, art, philosophy, or education while seeking Platonic romance with high aesthetic standards. The earth-shaped hand is broad, thick, irregular, yellowish, with thick firm muscles, wide wrist, and flat thick fingers. Owners are energetic, practical, confident, stubborn, and suited to inventor, explorer, or technician work; romance forms easily but often ends unresolved.

The ideal hand type features a small hand, long slender palm, oval fingers, fair soft skin, and almond nails; owners are sensitive, beauty-loving dreamers or artists but often impractical. The easygoing type has shorter, broader hands with tapering fingers; owners are artistic and impulsive yet prefer ease, with achievement depending on thumb firmness. Hands contain dense nerve endings. Six basic forms exist.

A firm handshake conveys proactive confidence; a weak one signals passivity. Equal handshakes feature full palm contact, moderate pressure, and matched head-spine posture. Dry palm and firm grip denote confidence, while weak or sweaty grips suggest weakness or anxiety. Standard equal handshake uses vertical palms with mutual pressure and three eye-contact glances. Politician-style uses both hands. Forearm or shoulder grasp suits close friends. Palm-down extension signals superiority; palm-up signals inferiority. Rotating the other's hand conveys dominance. Firm, strong handshakes convey excessive sincerity that often registers as insincere and domineering; such individuals are energetic, confident, arrogant, and possess strong leadership skills. Moderate-pressure handshakes with steady motion and direct eye contact signal resolute, frank, responsible, and reliable people who reason carefully and earn trust. Light-touch handshakes suggest pessimism, indifference, and a carefree attitude, yet also openness and modesty. Continuous up-and-down shaking marks extreme optimists. Prolonged holding without release indicates emotional richness and friendliness; between men it often signals a request, between women a tendency to gossip, and between opposite sexes romantic interest. Grasping only with fingers denotes sensitivity, emotional instability, kindness, and compassion. Reluctance to shake hands points to introversion, shyness, and sincerity. Palm-scratching during release reflects emotional depth and loyalty. Two-handed clasps, forearm or shoulder grips, and bear hugs are reserved for close friends. Flipping the partner's hand palm-down asserts dominance. Humble palm-up handshakes express respect or submission. Finger-grasp styles indicate reserve between sexes and coldness or status between same-sex pairs.

Charles Darwin observed that open palms held upward before the chest signal candor and submission, often paired with statements such as "I have nothing to hide." Placing both hands behind the back while walking or conversing projects high status and distance, known as the imperial stance. Baseball umpires cross arms over the chest or rest hands on hips to signal "You're out." Clenching a fist indicates solemn confidence during oaths, latent aggression, or resistance to new people and ideas. Finger tapping signals impatience or inward frustration. The steeple gesture—fingertips touching with palms apart—denotes confidence among high achievers such as lawyers, managers, professors, and negotiators; a lower steeple appears more in women. Steve McQueen's unconscious use in poker scenes illustrated it as a bluff tell.

Palms up during speech convey honesty; hidden palms suggest concealment. Hands on hips reflect impatience, readiness, dominance, or territorial defense, typical of police, soldiers, and leaders, varying by sex and context. Negative attitudes appear in clenched fists, thumbs pointed inward on thighs, fingers pressed to the forehead, or repeated table tapping. Freezing or constrained hand postures on stage indicate fear of rejection and low confidence. Fingers interlaced and clasped tightly under pressure show extreme insecurity; whitening knuckles mark rising tension. Ritual handshakes are brief and light. Pocket-watch carriers manage

time well and favor nostalgia; wristwatch style reflects attitudes toward time. Such individuals have conservative personalities marked by respect for authority, adherence to rules, and strong responsibility; they accept new ideas slowly and show emotional instability. Their strengths are patience and perseverance.

Resting the chin signals psychological boredom or listlessness. Carrying a briefcase in formal settings indicates a manager or executive who values efficiency and sets high standards; women who do so are typically assertive. Albert Mehrabian

## Voice and speech

Two politicians facing each other display contrasting signals: one leans back with legs crossed in a “4” shape and hands behind his head, projecting smug dominance, while the other hunches forward with limp hands, signaling defeat. The “4” posture conveys defensiveness, competitive arrogance, and negotiation impasse unless concessions occur. Hands clasped behind the back with one gripping the opposite wrist indicate extreme tension, frustration, and a need for self-control. Face-to-face seating creates distance and confrontation, suiting first meetings or business, whereas side-by-side seating fosters connection. Steady, grounded walking reflects composure; chin-raised, stiff-legged strides with swinging arms signal arrogance; brisk walking indicates efficiency and planning. An authoritative stance—hands behind the back, one gripping the other, body erect—projects power, used by officials inspecting sites or superiors addressing subordinates.

Confident standing places feet stably apart with relaxed knees and slight outward toe angle, paired with sustained eye contact. Inferior stance shows rounded shoulders, lowered head, and downward gaze. Natural seated thigh angle is roughly 30 degrees; wider angles claim dominance. Sudden foot freezing or inward-turned, interlocked toes signal insecurity. T-stance (one foot forward, weight on the other) reveals interest via front-foot direction. Leg crossing after a suggestion indicates disinterest; repeated shifts show growing impatience. Spread legs defend territory and assert dominance. Feet on a table or draped over a chair arm claim superiority. Pacing signals active problem-solving. Central seating preference marks self-centered individuals. Corner-walking reflects inferiority and eccentricity. Leg-rubbing downward relieves tension. Resolute seated posture (thighs apart, heels together, hands at navel) denotes courage and leadership but possessiveness. Shy seated posture (knees together, calves in “8” shape, palms between knees) marks conservatism.

Five basic standing postures exist: natural, T-stance, legs apart, legs crossed, and attention. Habitual leaning against objects while standing reflects candid acceptance yet lack of independence. Leg and foot shaking while seated indicates selfishness and strong but self-focused thinking; toe-tapping signals belief that victory is assured. Legs-apart standing posture occurs in 90 percent of cases among those with strong superiority and fighting spirit. Women walking hurriedly appear outgoing; large sturdy women walking with extreme caution prove shrewd. Feet rank as the most honest body parts, farthest from brain control. Leisurely sitting—half-reclining, hands clasped behind head—shows easygoing yet rigorous logic and perseverance. Confident sitting—left foot crossed over right, hands crossed at base—denotes natural intelligence and leadership.

Splay-footed walking splits into pigeon-toed (honest, detail-oriented) and duck-footed (bold, carefree). Dissolute sitting—legs spread wide—marks novelty-seeking and popularity. Crossed legs signal natural confidence; one leg hooked over the other adds caution. Turned feet in interrogation point toward the agent when relaxed and toward an exit when uneasy. Walking with hands clasped behind the back indicates preoccupation with difficult problems. Legs spread wide functions as territory defense under pressure. Hurried non-emergency walking with long strides shows abundant energy and integrity. Sitting upright and still with soles flat denotes sincerity and perfectionism. Feet together with toes lifted mark caution and sensitivity to

criticism. High heels correlate with performance desire and endurance of pain. Crossed-leg standing or sitting acts as defense signaling shyness. Gentle sitting—legs and heels tight, hands on knees—indicates introversion and humility. Toes together heels apart reflect over-seriousness and indecisiveness. Always wearing one fixed shoe style signals rigor and reliability. Walking with hands on hips denotes impatience. Foot movement shifting from side-to-side rocking to up-and-down kicking registers unconscious negative reaction. Chin-raised walking with exaggerated arm swing indicates self-satisfaction. Slightly leaning or hunched walking marks introversion and humility. Sleeping on one's back correlates with outgoing generosity and strong interpersonal relations.

People who stand with crossed legs or arms maintain greater interpersonal distance and signal reservation. Leg-shaking that appears suddenly and subtly reveals micro-reaction of discomfort. Walking styles convey distinct traits: head lowered with dragging feet and hands in pockets indicates depression; neat, orderly steps reflect strong willpower; steady, unhurried steps signal pragmatism; leaning forward suggests humility. The Lincoln-style posture—upright with legs apart, hands on armrests, jacket open—conveys frankness. The figure-four position with hands behind the head expresses satisfaction and control

## Part VIII — Applications

### Workplace and professional contexts

People who put others first earn trust through small conveniences. Examples include elevator mirrors for wheelchair users, Ye Shengtao teaching his son to offer a pen tip-first, and banker Lafeist, rejected fifty-two times, who retrieved a thumbtack and was hired by the witnessing chairman. Cheng Yi called this habit the highest form of learning.

Salespeople use forced-choice questions that presuppose purchase, such as "One egg or two?" which raised congee-shop sales above the direct query "Add an egg?" Similar traps include "Leather or fabric sofa?" and "Tuesday or Wednesday delivery?" Both options may still be refused.

Complaints about a third party often address the listener indirectly, the classic "point at the mulberry while cursing the locust tree." Mr. Ma cited poor soundproofing rather than his neighbor's late piano; the neighbor added padding. A department-store manager served a waiting customer himself, then recounted another employee's penalty to prompt Xiaoqiao without public rebuke. Soliciting a third party's opinion usually discloses the speaker's own view while disclaiming responsibility, a device repeated in *Dream of the Red Chamber* by Xifeng and Granny Lai.

Married women at gatherings form clusters, as do married men; singles seek other singles. Couples on dates or newlyweds stand close, touch frequently, and rest an arm around waist or neck. Stiff posture, forced rectangular smiles, and quick withdrawal from touch signal unresolved arguments; deliberate distance on the same sofa indicates ongoing tension.

Charles Schwab began as a stable hand at fifteen, studied construction texts at night, and rose to chairman of Carnegie Steel before founding Bethlehem Steel. Zhu Feng assumed equal status with the owner after rapid promotion; Zhang Fan volunteered for reports, then reassigned director Lao Guo's tasks and lost the subsequent election by a wide margin. A newly appointed Shandong magistrate who asked his superior's surname, banner affiliation, and province of origin was reassigned to school instructor in Guangxi.

A mold-making company posted exact process quotas and pay rates; workers needed no supervision. Mr. Chen arranged repeated "coincidental" meetings and a same-flight trip with Ms. Zhuo before showing her a

prepared office; she became his partner. An automobile salesperson matched a modest couple's need for reliable low-cost transport and a wealthy buyer's desire for status. Mei Ling met her future husband when he shared an umbrella; she later learned he had staged daily subway and elevator encounters from an office two stops away.

Dale Carnegie advised a manager to list tasks by priority and finish one at a time; the manager later cleared his desk and rose nationally. A psychologist heard three church builders answer "Who are you working for?" with resentment at stones, need for wages, or pride in building a city; only the third received greater responsibility.

Japanese wives in the 1980s signaled mood by flower-bouquet placement. Signs others are ready to speak include sustained eye contact, throat-clearing, closer distance, forward lean, open limbs, and smiling. Deliberate "coincidences" build trust yet can mask scams.

Humility draws helpers; Bill Gates annotated every speech draft with staff. Career analysis of 16,000 people showed failures often stem from mismatched roles. Mr. Davord succeeded in mechanical manufacturing after leaving accounting. Three questions before changing jobs: what dissatisfies the current role, risks of the new one, and contingency plans.

FBI experiments found 56 percent of informants agreed to help in a food-filled room versus 20 percent in an empty room. Agents raise their chair at least 5 cm above the interviewee's or use wheeled chairs to adjust distance. Jesse elicits nods with three opening confirmations, then nods noticeably; Matarazzo's experiment showed vigorous nodding increased speaking time in seventeen of twenty participants. Doubling the usual nodding rate doubles perceived agreement.

In purposeful exchanges, interruption exerts control; FBI agent Peter repeatedly broke Zorro's monologues by requesting term definitions or examples. Dutch psychologist Arco Kalma states that successful interruption increases power.

Wide-open eyes are preferred; men viewing bathing photographs show 11.3 percent average pupil dilation and swimsuit images 20 percent. Consciously widening the eyes produces two to three times the impression of normal eyes. Standing conveys greater presence; 59 percent of viewers rated a standing figure more imposing. FBI supervisor Matt always stands during meetings. Requests for promotion succeed more often when the requester stands and the decision-maker sits.

When late, say

## Part IX — Quizzes

### Emotion Stability Quiz

Expression recognition tests are structured in three tiers of 6–8 images each. Beginner tests cover the six basic emotions—happy smile, surprise, disgust, anger, fear, sadness—each scored at 5 points. Intermediate tests examine smile sincerity, surprise intensity, disdain, mild anger, worry, and melancholy, stressing eyebrow, eyelid, cheek, and mouth morphology plus force-matching among features. Advanced tests address blended states such as suppressed disgust, unconvinced disagreement, startle mixed with sadness or fear, and smug or pleading smiles, using cues like upper-eyelid droop to the pupil edge or chin wrinkling.

Scoring ranges are 0–50 for recognizing obvious expressions only, 50–60 for good intuition but self-doubt, 60–80 for solid ability, and above 80 for consistent judgment in cases. Psychological testing targets stress

micro-expressions for lie detection after filtering disguises. Real-life micro-expressions consist of subtle incomplete brief muscle movements that leak under extreme suppression and compound expressions blending multiple emotions.

Beginner Test 1 reference answer is happy smile. Section 1.1 covers recognition of ordinary expressions via six images; each correct intuitive judgment close in meaning earns 5 points. Beginner Test 2 reference answer is surprise, marked by widened eyes and mouth

## Part X — Peripheral topics

### Personality types and patterns

A breakup dialogue can trigger surprise at an unexpected statement, disgust at an ex-boyfriend's call, anger at an implied threat, fear of irreversible loss, and sadness once the outcome is final. Paul Ekman identified six universal emotions—surprise, disgust, anger, fear, sadness, and pleasure—that transcend culture; his New Guinea fieldwork with isolated tribes confirmed the finding. Negative stimuli escalate from disgust (including contempt) through anger, fear, and sadness; positive stimuli produce pleasure.

Men exhibit darker skin, coarser hair, and greater muscle mass; women show lighter skin, finer hair, and softer build. Two surveys found more than 75 percent of successful career women reported unsatisfactory emotional lives, while 90 percent of ordinary unmarried men preferred a dependent future wife. Women who never wear makeup fall into three groups: those too occupied with survival, those raised to avoid it, and those who deliberately pursue natural appearance. Bright-red lipstick signals an outgoing personality suited to public-relations work; pink indicates gentleness; purple reflects a desire for attention.

The Barnum effect leads people to accept vague personality descriptions as personally accurate, as illustrated by Einstein's father's chimney-soot anecdote. Avoidance requires facing oneself directly, gathering sufficient information before judging, and remaining calm. In romantic contexts, individuals conceal flaws until daily life reveals them, often lowering prior compatibility ratings from 90 to 60 points. "Love personality" creates illusions of perfection; couples often face marital unhappiness once daily life reveals character and habit differences, as illustrated by a woman reconsidering her view of Mr. Reginald versus her husband after seven or eight years.

Parents forcing an eleven-year-old bookish boy into football, or a slight European tailor into mechanical work, ignore innate types: intellectual (large head, high forehead, logical), energetic (plump, inner reserves), active (lively, extroverted). Mixed types include intellectual-active (Henry Ford, Madame Curie, General Pershing), intellectual-energetic (Nicholas Murray Butler, Hendrik Willem van Loon, Orson Welles), and active-energetic (Mussolini, Andrew Carnegie, Charles Schwab). Match tasks accordingly.

Watches reveal traits: electronic (independent, mysterious), LCD (frugal), alarm (disciplined), multi-zone (unrealistic), gold classic (long-term planners), pocket (balanced), wind-up (self-reliant), numeral-free (abstract), designer (image-conscious), none (autonomous). Handbag choices indicate: ordinary (conformist), striking (individual or attention-seeking), casual (life-enjoying), briefcase (executive, cautious), small decorative (fragile), medium shoulder (independent yet conservative), ethnic (out of place), oversized (low responsibility), many pockets (orderly), metal (trendy), masculine (strong, extroverted), neutral (non-confrontational). Payment styles: in-person (traditional, approval-seeking), delayed (selfish), shifted (irresponsible), immediate (decisive), telephone (new-tech reliant).

Accessories include gold earrings (confident), silver (rule-bound), old-style (family-loyal), large (attention-seeking), gem-set (appearance-focused); bracelets (energetic, idealistic); ornate wedding rings (notice-seeking), crested (family-valuing), diamond (vain), complex (strict, stubborn). Bedroom use signals: multi-purpose (outgoing), sleep-only (orderly), celebrity posters (withdrawn, insecure), lavish but styleless (convention-bound), toys/fitness gear (childlike, passionate), messy yet presentable living room (sluggish), childhood items retained (nostalgic, immature). Marriage requires balancing similarities and differences; partners should share interests, complement strengths, and accept shortcomings.

Everyone has a distinct scent that mirrors romantic personality. A quick test links choices to traits: rose garden signals selfless passion; Far East harbor indicates mysterious observation; desert flowers suggest independent melancholy; internet café reflects novelty-seeking; hot spring denotes pure appreciation of beauty. Fosdick noted marriages fail when people do. Advice includes matching temperaments by build, seeking fuller occiputs for loyalty, wider temples for financial skill, high foreheads or chins for perseverance, and avoiding thick lips, fleshy napes, or heavy eyelids signaling greed.

Beard density and style reveal male character: thick beards mark outgoing trustworthiness; sparse ones suggest refinement or passivity; screw-curved beards indicate creativity and leadership; long beards show fleeting romance; powerful beards denote

## Defense mechanisms

Dr. Yang confronted a patient overheard criticizing medical staff, only to learn the remarks targeted nurses. The patient offered no resistance, so Yang later scolded the nurses, who cited the patient's demanding behavior amid heavy workloads. Old Wei hosted a seemingly wealthy Hong Kong merchant who left subordinates with an unpaid hotel bill after feigning drunkenness; investigation showed the man was deeply indebted and seeking favors under false pretenses, leading to police intervention.

Books kept at work reveal attitudes. Financial magazines signal competitiveness, authority reverence, and opportunity spotting. Fashion magazines indicate trend focus and appearance orientation. Martial-arts novels point to imagination, romance, and emotional delicacy. Romance novels suggest emotional attunement, insight, confidence, and open-mindedness. Biographies mark curiosity, caution, and big-picture oversight. Comic books reflect playfulness, outgoing nature, and later realization of naivety. Detective novels denote mental challenge and multi-angle problem solving. Horror novels show stimulation seeking and complexity tolerance. Science-fiction novels indicate imagination, technology attraction, future planning, and inconsistency.

Dual personality denotes two stable, opposing modes of thinking and acting that coexist and shape behavior regardless of context. Film character Travis exemplifies this through simultaneous degeneracy and heroic impulses, producing erratic actions under stress. Such individuals display poor social adaptation, self-righteousness, and defensive rationalization.

Hobbies expose core nature:

- Performing signals delicate emotions and imagination.
- Fishing indicates process orientation and purity seeking.
- Gardening reflects life enjoyment and sensitivity.
- Gourmet cooking marks creativity and goal persistence.
- Chess or cards suggest intellectual superiority and logical strength.
- Writing denotes analytical strength and insight seeking.

- Stamp collecting points to self-regulation and face consciousness.
- Hunting reflects ruggedness, loyalty, and courage.
- Traveling indicates outgoing curiosity and stimulation seeking.
- High-risk activities mark robustness, meticulousness, and decisiveness.

Desk organization displays character. A neat surface and drawers signal efficiency, self-discipline, and time valuing. A tidy surface with chaotic drawers indicates appearance focus and superficial sociability. Both empty suggest impatience and ambition. Both messy reflect unplanned haste and detail disregard. Mementos present mark introversion and emotional richness. Cash stored inside signals suspicion and theft worry.

Fat people typically exhibit vitality, calm security, affability, humor, and leadership through delegation, though they resist exercise. Thin people appear sensitive, tense, alert, and perfection-seeking yet emotionally unstable. Slender types pursue refined beauty and elegant settings.

Judging character from bodily features requires caution. Dr. Bai faced a woman in her thirties whose stomach-pain complaint veered into antidepressant confusion, neuralgia, and repeated topic shifts. German psychiatrist Ernst Kretschmer linked body type to personality through four types—pyknic, asthenic, athletic, and dysplastic—and differing disorder susceptibilities. Robust types show large build, coarse stiff hair, and deep voice; they are rough, bold, impulsive, and physically powerful, yet may prove intelligent and imaginative, as with sculptor Rodin or mayor Fiorello La Guardia. They ignore details, favor blunt speech, and display leadership, as Lenin did.

Skull shape correlates with personality. Wide-shaped heads (6.5 inches or more at ear tops) indicate energy, aggression, and greed. High-shaped heads (5.8 inches or more from ears to crown) mark strong will and ambition, common in educators, politicians, and judges. Long-shaped heads (7.6 inches or more front to back) signal future orientation and sociability. Short-shaped heads (under 7.6 inches) reflect self-focus and risk tolerance if rounded, or caution if square. Low-shaped heads (under 5.8 inches from ear tops to crown) belong to pragmatists focused on material ideals. Narrow-shaped heads (under 6.5 inches wide) indicate born diplomats who are modest and patient. Round heads are impulsive yet adaptable; square heads are excessively cautious, as when square-headed crane operators caused repeated accidents while round-headed Michael operated safely.

Passive-aggressive personalities express anger through negativity, procrastination, and errors while in weaker positions. Paranoid personalities display five traits per the Chinese Classification of Mental Disorders: interpreting neutral acts as malicious, extreme conceit, pathological jealousy, self-referential delusions, and unfounded infidelity suspicions. They remain suspicious and rigid due to early lack of love or later setbacks; interact with tolerance yet firm distance. Histrionic personalities perform dramatically for attention, dress flamboyantly, prove highly suggestible, and shift emotions

## Manipulation modes

People who like drawing straight lines or crosses overflow with vitality, dislike solitude, and stir trouble. Histrionic personalities fantasize, treat imagination as reality, and exaggerate feelings. Sitting and sleeping with hands at sides and legs straight reflects constant high tension. Dark-green preference signals ideals tempered by worldliness. Window-seat choice marks ordinary personalities that avoid noise. Ballet enthusiasts display strong patience and discipline. Foreign-travel fans are fashionable and confident about change. Comedy lovers appear casual yet harbor strong emotions once attached. Crossed arms and legs in sleep indicate strong self-defense alongside fragile character.

Large, flourished handwriting often masks insecurity. Continuous circular drawings mark empathetic, trustworthy, easily satisfied people. Outdoor camping travelers show independence, imagination, and attention to objective reality. High-and-narrow foreheads excel at analysis but prove critical and narrow-minded. Pop-music fans pursue simple, relaxed lifestyles. Small sawtooth patterns signal mental agility and quick analysis. Low-in-front, high-in-back head shapes are arrogant, willful, and poor judges of character. Rock dancing appeals to the young, avant-garde, and rebellious. Classical-music listeners are strongly rational and reflective.

Sitting with back to a wall signals aloofness. High-in-front, low-in-back heads are highly compassionate, sometimes excessively so. Wide-and-low foreheads are versatile yet scatter mental energy. Latin-dance enthusiasts are energetic and performance-driven. Social dancing indicates interaction skills and organizational ability. Travel-souvenir collectors pursue novel things. Talk-show viewers have active minds and rich imaginations.

Narcissistic personalities demand special treatment, exaggerate achievements, flatter superiors while envying them, and discard people once usefulness ends. Sakuragi Hanamichi in Slam Dunk boasts of genius status and misreads rivals' disdain as jealousy. Manager Li treats juniors with contempt and engages only with superiors. Childhood influences often produce these cold traits; one man blamed his ill friend for a missed interview rather than showing concern.

Obsessive-compulsive personalities appear in rigid "workaholics." Yi Sun-sin exemplified this through meticulous planning that aided wartime success yet strained peacetime relations. Director Ma fixated on minor details after decades of overtime. Manifestations include excessive focus on lists, reluctance to delegate, and retention of childhood items. Strict parenting often contributes. Mild forms aided Stanley Kubrick and Stephen Chow.

Compensation redirects suppressed anxiety to safer targets, as when a father slapped himself after his son was beaten or a chain of office scolding ended with a cleaner kicking her dog. Wu Lili transferred affection from Sun Gang to a classmate sharing minor traits. Sweet-lemon reframing calls an overweight partner preferable to a thin one. Lapse occurs when reason fails, producing slips such as Han Qiaosheng's "Fan Zhiyi ran a high fever—36.8 °C." Evaluator Yang deduced Zhang Chao's guilt from the slip "called for Madam Zhao San, not Zhao San himself." Zhang Zuolin wrote "hand-black" instead of "hand-brushed ink."

Daydreaming discharges desires through fantasy; Xiao Hua imagined superpowers against extortionists. Freud linked artistic creation to sustained daydreaming, as seen in YY novels and superhero narratives. A Qing-era woman's Dream of the Red Chamber addiction required burning all copies. Counter-attitude produces exaggerated compliance that masks resentment; Xiao Wang offered a bribe then spat once the officer left. Old Lady Pang performed grandmotherly affection in public yet treated grandchildren coldly in private.

Projection attributes forbidden impulses to others; Widow Li spread rumors about Widow Zhang after her own rejection. A capital official prepared one hundred tall hats yet denounced flattery until told few were upright. Displacement redirects emotion; a wife angered by her mother-in-law scolds her husband instead. A migrant millionaire cut Armani suits after a clerk's insult. Sublimation channels impulses into valued work; Schubert produced over one thousand pieces amid heartbreak, and Qu Yuan, Li Bai, Du Fu, and Li Yu followed similar patterns.

Cognitive dissonance intensifies proselytizing when predictions fail, as with Marian Keech's followers. Smokers supply justifications after accepting a cigarette. William James noted that calmly accepting unchangeable facts is the first step past misfortune. Confucian tales and Bill Gates's fire-salvage metaphor illustrate learning from

mistakes. Overgeneralization turns one setback into total self-condemnation. Mislabeling applies global negative tags that become self-fulfilling. Contrasting cancer cases show autosuggestion effects; Émile Coué's phrase "Every day in every way I am getting better"

## Psychological games and effects

A psychologist's worry-box experiment found that 90 percent of anticipated concerns never materialized and most of the remaining 10 percent resolved without intervention. Research shows 40 percent of worries concern the past and 50 percent the future, with 92 percent never occurring and the remaining 8 percent readily managed. Anxiety comprises four intensity levels: bodily tension with inability to relax; autonomic responses such as sweating, tachycardia, and gastrointestinal distress; unwarranted future worry; and excessive vigilance to minor stimuli. Elderly hospital patients sometimes refuse discharge and exhibit regressive behavior that seeks the institution as a childhood-like sanctuary amid feelings of powerlessness.

The Johari Window divides the self into four panes: Open Self (public facts), Blind Self (unknown to self), Hidden Self (private information), and Unknown Self (latent potential and subconscious traits). Behavioral psychology aids exploration of the Unknown Self. Cognitive dissonance is reduced by changing a cognition, adding a new cognition, altering the importance of a cognition, or changing behavior. Negative self-suggestions include "I will definitely mess this up," "I need help," and "My idea is definitely wrong." Life-event stress scores above 300 points correlate with 89 percent of individuals developing despair and related illnesses. Jealousy arises only within the same domain when superiority is lost. Primacy-effect studies show initial information disproportionately shapes social judgments. Suspicion is countered by rational examination of evidence, recognition of personal strengths, regular communication, vertical rather than horizontal comparison, and cultivation of small achievements through prioritized task lists.

Everyone possesses unique defense mechanisms that function as psychological transformers, preventing loss of control during anger, despair, or overexcitement. Anxiety often arises from time pressure; the remedy is closing the eyes, breathing deeply, and settling the mind before focusing on the task. Negative emotions stem primarily from flawed thinking patterns. Everyone has strengths that, once identified, should be displayed boldly. Jealousy can be vented to relatives and friends or transformed into motivation by perfecting oneself. The regression mechanism offers only temporary relief. The attack mechanism provides direct violent, soft, or cold catharsis but risks severe consequences.

The liking principle states that people comply more readily with requests from those they know and like, known as the "one of us effect." Application involves enhancing physical attractiveness, identifying similarities, and increasing contact. Wang Le won clients by showing she liked them. Howard Thurston's remark "I sincerely love my audience" and Elliot Aronson's survey illustrate that liking begets liking. The good-cop/bad-cop technique exploits this by creating contrast. The reciprocity principle requires returning favors. Examples include an office director who absorbed blame and later received advancement, a car salesperson whose soiled handkerchief prompted a purchase, strangers replying to unsolicited Christmas cards, a World War I German soldier who spared an enemy after receiving bread, and disabled-veterans mailings that included address labels and raised donations from 18 percent to 35 percent.

Conformity leads people to match group behavior. In a Candid Camera episode, a man removed his outer clothes after seeing others in underwear. Canned laughter increases viewer laughter. Sony marketed the Walkman by having employees demonstrate it visibly near schools. Authority produces blind obedience. In Stanley Milgram's 1961 Yale experiment, 40 volunteers administered shocks under orders; two-thirds continued despite protests. Aviation data show the co-pilot's reluctance to correct the captain contributed to

20 percent of fatal crashes. A 1993 Northwest Airlines flight crashed after the co-pilot only asked deferentially about altitude.

Commitment-consistency pressures people to align actions with prior statements. Racetrack bettors' confidence rises immediately after wagering. Written and public commitments strengthen follow-through. Scarcity intensifies desire. Merchants use limited-time sales and "others are interested" statements. In negotiations, creating the illusion of alternatives forces concessions.

Traumatic one-time learning produces lasting compliance. Zhang Li's public outburst over a round bed costing 18,000 yuan reversed household authority. Intermittent reinforcement creates stronger persistence than continuous reward. Pigeon B, rewarded irregularly, continued lever-pecking far longer than Pigeon A after rewards ceased. Positive reinforcement builds loyalty through praise. Negative reinforcement works faster: a mouse shocked in a black compartment learned to flee to the white compartment in seconds. Punishment deters through fear. Manager Fang retained Xiao Hua by overloading her with tasks and withholding conversion confirmation. Project manager punished Ai Li with desk-slamming tirades after a missed call.

The cake-division effect occurs in zero

## Appendix A — Sources

The dataset catalogs nine digital items by stem. Stem 9786267173633 measures 173333 bytes with SHA256 46ea3c29f3fc94a11f8edcbc7ece3f04aaa573c560ebb8641572d804339bdfd5. Stem 9787510438820 holds 593626 bytes and hash 236e16b6a657a87431ed1213868dfd2207c66a73ad09be26b98e3f73d82f5d2a. Stem 9787519023317 is 185314 bytes with hash 2bbfcd3b258d3f659ccdc58abc22dd45b4be5b5f8c3bef0f61a41983b51175b3. Stem 9787536066236 contains 244248 bytes and hash 300073d9c28983105fb2909532f22f9e1904c8045f954b36b8beb8a268557417. Stem 9787538583373 measures 192056 bytes with hash edc170111ab3139131e640f7dbd89c5b3a889fed147b4d3a1fc82631a6db507f. Stem 9787554606841 reaches 776940 bytes and hash b293855c4b09e8b3a95f6f902d8312f09ed5873fecca5100f4424f63c5eb3df8.

- Stem 978

## Appendix B — Quizzes

Psychological assessments of emotional stability rely on self-reported responses. One evaluation uses 30 items on morning mood, social feedback, self-image, checking behaviors, future worries, family satisfaction, perceived abilities, unusual perceptions, paranoia, suicide views, unease from others, nighttime fears, family treatment, recurring dreams, nightmares, invisible worlds, parental doubts, feeling loved, temper, understanding, seasonal feelings, loneliness, withdrawal, resignation urges, intolerance, reactions to gossip, fallen flowers, public criticism, and post-argument responses. Scoring awards 2 points for A, 0 for B, and 1 for C. Scores of 0–20 denote stable emotions and strong self-confidence, 21–40 indicate basic stability with fluctuating enthusiasm, and above 40 signal emotional instability.

A second test measures emotion control via 10 items on reactions to rain while driving, bicycle collisions, being yelled at, annoyance, lost clothing, illness-related rest, happiness, job searches, friend dissatisfaction, and causing friend unhappiness. It scores 1 point for A, 2 for B, and 3 for C. Totals of 1–10 reflect weak control, 11–20 average ability, and 21–30 strong control.

Inner fragility is assessed with one item on reactions to tripping over an iron bar: arguing signals extreme fragility with escape tendencies, claiming compensation shows apparent strength masking low moods, and accepting bad luck indicates resilience.

An emotion type test employs 8 items on challenging emotions, change attitudes, preferences, physical symptoms, management strengths, emotion traits, intolerances, and forced choices among mission, prediction, or optimism. Most A selections identify the time bomb type, most B the people pleaser type, and most C the emotionally depleted type.

Irritability assessment covers 25 situations including faulty devices, extortion, selective corrections, stuck vehicles, ignored speech, deception, spilled coffee, knocked clothing, followed shoppers, canceled plans, teasing, honking, parking insults, blame-shifting, foot-tapping, unreturned items, forgotten tasks, blocked expression, off-topic insistence, interruptions, slow drivers, gum, group laughter, torn pants, and lost coins. Totals of 0–45 indicate low anger, 46–55 above-average calm, 56–75 the normal range, 76–85 easier anger, and 86–100 frequent loss of control.

Depression indexing rates 15 items on sadness, hopelessness, worthlessness, inferiority, guilt, indecision, irritability, lost interest, lack of motivation, aging perceptions, appetite shifts, sleep issues, reduced libido, health worries, and suicidal impulses. Totals of 0–4 indicate no or mild depression, 5–10 normal but unhappy states, 11–20 moderate depression approaching, 21–30 moderate depression confirmed, and 31–45 severe depression.

The Burns Anxiety Scale lists 33 symptoms including anxiety, derealization, depersonalization, panic, death fear, tension, concentration difficulty, disordered thinking, frightening fantasies, loss of control, breakdown worries, fainting fears, heart attack concerns, foolishness fears, abandonment fears, rejection fears, impending doom, rapid heartbeat, chest pain,

## Appendix D — Bibliography

Foundational works in nonverbal communication include Alfred Adler's *Understanding Human Nature*, Gordon Allport's *Studies in Expressive Movement*, Michael Argyle's *The Psychology of Interpersonal Behaviour*, Ray Birdwhistell's *Kinesics and Context*, Charles Darwin's *The Expression of the Emotions in Man and Animals* and *On the Origin of Species*, Julius Fast's *Body Language*, Erving Goffman's *Behavior in Public Places* and *Interaction Ritual*, Edward Hall's *The Hidden Dimension*, Konrad Lorenz's *King Solomon's Ring* and *On Aggression*, Abraham Maslow's *Motivation and Personality*, and Gerard Nierenberg and Henry Calero's *The New Art of Negotiating*. Later sources add Paul Ekman's research on facial expressions and deception, Joe Navarro and Marvin Karlins' *What Every BODY Is Saying*, Allan and Barbara Pease's *The Definitive Book of Body Language*, and Gray's *Anatomy* (39th edition).

Additional references include *A Treatise of Human Nature*, *The Descent of Man and Selection in Relation to Sex*, *The Principles of Psychology*, *Games People Play*, *Detecting Lies and Deceit*, *Telling Lies*, *Emotions Revealed*, *How to Read a Person Like a Book*, *Evolutionary Psychology* (2nd ed.), *Social Psychology* (8th ed.), *Personality Psychology* (4th ed.), and *The Psychology of Criminal Conduct*. Journals cited include *Journal of Personality and Social Psychology*, *Psychological Monographs*, and *Experimental Psychology*. Anatomical sources draw from *Neuropsychology*, *Physiological Psychology*, and *Medical Psychology*.

Key indexed attitudes and signals cover recognition, offense, anger, resentment, boredom, confidence, cooperation, resistance, disgust, superiority, doubt, worry, expectation, fear, unease, frustration, happiness,

surprise, interest, admiration, anxiety, openness, willingness to converse, readiness, self-affirmation, courtship, satisfaction, self-control, contentment, shyness, and concealment.

- Specific gestures and postures include crossed arms, arm swing while walking, chin stroking, lip biting, blink rate, rapid breathing, clenched fists, clenched jaw, throat clearing, scalp scratching, finger tapping, sideways glance, pupil dilation, furrowed brow, hands steepled, thumbs up, open palms, sweaty palms, ankles crossed, legs crossed in “four” posture, feet on table, one leg draped over chair arm, Lincoln-style sitting, head lowered or held high, head tilted, shoulders drooping, hair smoothing, disheveled hair, nail biting